

Singh Sanskar Surendrasingh

DSIRe Lab, Block-3, IIT Delhi

+91 7984399359

ssanskar2705@gmail.com

sanskar2705.github.io/Portfolio

sanskar2705

Research Interests

My research interests span information retrieval and multimodal representation learning, with a diagnostic focus on understanding system behaviour through the geometry of learned representations. I am increasingly interested in NLP and mechanistic interpretability, particularly in how information is encoded in model internals and how this structure shapes model behaviour.

Education

Indian Institute of Technology (IIT) Kharagpur

CGPA: 8.46 / 10

Master of Technology in Wireless Communication & Networks

Jun 2023 – May 2025

- Master's Thesis: *Facial Recognition and Verification using a Vision-Transformer Siamese Network* (advised by Prof. Saumik Bhattacharya & Prof. Suva Sekhar Das).

Gujarat Technological University

CGPA: 7.83 / 10

Bachelor of Engineering in Electrical Engineering

Jun 2019 – May 2023

Publications

- Scalability Analysis of Cross-Modal Text-to-Image Pipelines.**
Sanskar S. Singh, Ziv Barretto, Bharati Khanijo, Aditya Sahu, Maya Ramanath.
CIKM 2026 – Full Research Paper (*Accepted*).
- Retrieval-Stage Interactions in Cross-Modal Text-to-Image Retrieval Pipelines.**
Sanskar S. Singh, Ziv Barretto, Bharati Khanijo, Aditya Sahu, Maya Ramanath.
CIKM 2026 – Short Paper (*Accepted*).
- SetCoRe: Set Composition for Retrieval of Educational Explanations.**
Riti Verma, Sanskar S. Singh, Maya Ramanath.
ECIR 2027 (*under review*).
- Set2Story: Retrieval-Grounded Storyboard Construction for Educational Explanation.**
Aryan Ahmed, Sanskar S. Singh, Maya Ramanath.
ECIR 2027 (*under review*).
- SARCH: Multimodal Search for Archaeological Archives.**
N. Sinha, B. Khanijo, Sanskar S. Singh (3rd author), *et al.*
CIKM'25 MGenSR Workshop, Seoul – *Published*.

Research Experience

IIT Delhi · Advisor: Prof. Maya Ramanath

New Delhi, India

Project Scientist – Multimodal Information Retrieval, DSIRe Lab

May 2025 – Present

- Cross-modal retrieval pipelines at scale (first author; full paper accepted, CIKM 2026).** Systems-level study of single-index, retrieve-and-rerank, and ensemble pipelines on MS-COCO, Flickr30K, and a 7M-image LAION subset, measuring accuracy, latency, energy, storage, and candidate fidelity. Identified **O-IVFPQ as the strongest single-index trade-off**, with reranking dominating end-to-end cost once applied, making index choice secondary.
- Retrieval-stage interactions in cross-modal retrieval (first author; short paper accepted, CIKM 2026).** Studied how candidate generation, reranking, and fusion interact across retrieval pipelines on MS-COCO and Flickr30K. Found that **reranking gains depend on first-stage retriever quality**, candidate-set size helps reranking but hurts RRF ensembles, and ensemble gains stem from complementary query coverage, not raw accuracy.
- Spatial fidelity maps – query-time prediction of ANN retrieval quality** Candidate fidelity can only be measured by running the exact-KNN search an approximate index exists to avoid; proposed an **offline-built landmark lookup** that predicts per-query CF@100 from a query's nearest landmark alone, with no exact reference at query time. On a **1M-pair LAION pilot** (50K held-out queries; 9 HNSW/IVFPQ/O-IVFPQ configurations), it beat both a global-average and a density-only predictor in **all 16 model-modality-direction pairings** (4 embedding models × 4 query/index directions).
- SetCoRe (co-author; under review, ECIR 2027).** Built the first benchmark for **multimodal explanation-set retrieval**, pairing 440 learner queries with human-validated gold sets across 16 science textbooks; found that diversity-based set selection underperforms simple top-ranked retrieval.
- Set2Story (co-author; under review).** Formalized storyboard planning as the missing step between explanation-set retrieval and educational content generation, with **grounded resource selection, pedagogical ordering, role assignment, and time budgeting**; developed a benchmark, intrinsic evaluation metrics, and a lightweight grounded planning method.
- MMIR-Explorer:** designed and deployed an interactive platform for query-level analysis of retrieval-stage interactions – side-by-side pipeline comparison, reranking effects, candidate provenance, and ANN fidelity. Live at mmir.iitd.ac.in.
- SARCH – multimodal archival search (published, CIKM'25 Workshop):** co-developed a hybrid retrieval system over 296 scanned archaeology documents (63K+ pages) using modality-specific embeddings (MiniLM, CLIP, TAPAS) fused with BM25 via reciprocal-rank fusion.

Projects

IIT Kharagpur

One-Shot Facial Verification with a Siamese Network (Master's Thesis)

Jun 2024 – May 2025

- Trained a **Vision-Transformer Siamese model** on Celeb-DF v2 (31K+ images) for one-shot recognition and verification, using balanced (anchor, positive, negative) triplets, a metric-learning setup that mirrors learned similarity in dense retrieval.
- Achieved **94.83% recognition accuracy** and **91.06% verification accuracy** (98.44% precision, 85.77% recall).

Industry Experience

Eternal Robotics

Ahmedabad, India

Computer Vision Intern

Feb 2025 – May 2025

- Integrated and deployed a YOLO-based quality-inspection system for car assembly lines, and built an OCR proof-of-concept on vehicle stickers with Qwen2.5-VL-Instruct (**94.74% accuracy**); contributed to a CI/CD pipeline for model deployment.

Unizoy

Surat, India

Machine Learning Engineering Intern

May 2024 – Jul 2024

- Developed a real-time pose-estimation and feedback system (Mediapipe) covering 10+ exercises for remote physiotherapy, reaching **93% accuracy** on the biceps-curl model.

Freelance

Outlier AI

Remote

Mathematics & Language RLHF Expert

Jun 2024 – May 2025

- Domain expert evaluating and correcting large-language-model outputs for RLHF, on mathematics and regional-language reasoning projects.

Leadership & Mentoring

IIT Delhi

New Delhi, India

Research Mentor / Team Lead, DSIR Lab

2025 – Present

- Lead and mentor a team of B.Tech students and research interns in the lab, guiding them on multimodal retrieval and computer-vision research, experiment design, and scientific writing.

Ideathon, SSGEC Bhavnagar

Bhavnagar, India

Team Leader — 1st Place

Feb 2022

- Won 1st place for proposing a renewable energy-storage solution that harnesses kinetic energy from flowing river water and converts it to stored potential energy.

Teaching Experience

G.S. Sanyal School of Telecommunications, IIT Kharagpur

Kharagpur, India

Teaching Assistant — Communications Simulation Lab & Embedded Communications Lab

Jul 2024 – May 2025

- Mentored 20 students through simulation and embedded-systems lab projects, with hands-on demonstrations and feedback.

Skills

Languages: Python

ML / Deep Learning: PyTorch, TensorFlow, Keras, Hugging Face Transformers

Retrieval & IR: FAISS (KNN / HNSW / IVFPQ / O-IVFPQ), Apache Solr, BM25

Multimodal Models: CLIP, OpenCLIP, FLAVA, UniIR, BLIP-2, Qwen-VL, MiniLM

Tools & Libraries: Git, FastAPI, Flask, Streamlit, NumPy, pandas, Matplotlib, OpenCV, Linux

Spoken Languages: English (full professional), Hindi (native)